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Using Active Learning to Guide Price Optimisation

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Abstract

Setting an optimal price to achieve businesses' goals generally involves understanding how demand reacts to change in price ("price elasticity"), which can be a costly process, particularly for products with limited variability in historical prices and demand data since traditional approaches require an extensive amount of information. This paper investigates a potential approach to alleviate the costs associated with price experimenting for businesses, thereby facilitate their understanding of the price-demand relationship and assist in improving price optimisation. Through the application of Active Learning (AL) techniques such as Uncertainty sampling and Greedy sampling, a demand prediction model can selectively identify the most informative price points to learn, focusing on its potential to reduce data collection while maintaining model accuracy. Key findings indicate that models guided by AL strategies can rival the accuracy and optimisation capabilities of a model which randomly selects data points with up to 30% fewer data points. This reduction in data requirements translates to significantly lower costs, faster model deployment and more optimised price selection, thus offering businesses an advantage in rapidly changing markets and improved resources allocation to strategic initiatives. Furthermore, the study highlights the varying effectiveness of different AL techniques, suggesting that thorough understanding of the AL methods and the problem context should be obtained before implementation to outperform simpler approaches. Additionally, the implications for business decision-making are profound, with AL-driven models enabling more efficient and responsive pricing strategies. The study concludes with suggestions for future research to explore several limitations, including the addition of prediction factors and dynamic cost structure, to justify AL's broader applications across different problems.

1. Introduction

1.1. Price elasticity and its vital role in price optimisation

Price optimisation is a critical aspect of strategic management that aims to obtain a price strategy which maximises the revenue or profit whilst considering other factors including costs, competitors, etc. Especially, the interest in this topic becomes prevalent among businesses when future demand is uncertain according to Liu and Sustik (2021).

Price elasticity, which measures "the percentage changes in purchased quantity or demand with a 1% change in price" (Andreyeva et al., 2010), emerges as a viable solution to establish a connection between product price and sales quantity. Henceforth, research has concentrated on deriving accurate price-demand curves to facilitate the optimisation, namely the analysis of price elasticity of flight tickets (Abdella et al., 2019). Such information on price-demand responsiveness was used by airlines to dynamically determine the ticket charges or when to launch promotions to improve the overall demand.

Numerous models and methods were introduced to determine an optimal price to maximise profitability, namely the approach for e-commerce businesses mentioned by Funk (2011). Alternatively, retailers might utilise the relationship between price and demand to derive a desirable product mix for promotion campaigns and optimise the profit, referring to the work of Greenstein-Messica and Rokach (2020). Nevertheless, this paper will focus on single product pricing and how businesses can enhance such procedure, and deeper insights into empirical price optimisation efforts via price elasticity will be discussed in later sections.

Multiple challenges have been identified whilst attempting to model price elasticity. In this paper, we concentrate on tackling the lack of price variation in historical data. In many cases, it could be extremely costly for businesses to experiment altering product prices and gathering data on fluctuation in demands to derive an accurate demand curve. If the price is set higher than the customers' expectation, the demand might plunge greatly, combined with increased competition, since consumers becomes especially risk averse as studied by Heidhues and Kőszegi (2008). Conversely, decreasing price would diminish the profit margin, thus risking that the corresponding uplift in quantity sold might not compensate. Additionally, Morlotti et al. (2023) concluded consumers' decision to buy, wait or give up purchasing is substantially dependent on the prices presented to them over time. Thus, whether the prices are raised or lowered, companies will be jeopardising a significant proportion of their profitability to obtain demand data at different price points, not to mention the potential damage on customer loyalty or brand image.

Although prioritising price stability has its merits, this implies companies tend to be restricted to a small set of data to compute price elasticity. Henceforth, the developed optimisation models

are likely to face overfitting issues. Moreover, the accuracy of the demand curve and sales forecasts will depend on the price points at which demand is collected, which, in many cases, are randomly selected. Therefore, this scenario poses a considerable hurdle for statistical models that normally require a broad spectrum of information to provide reliable price elasticity and corresponding optimisation strategies. To address this challenge, we search for innovative approaches beyond traditional methods that can assist in plotting an accurate demand curve with fewer price points.

1.2. Paper objective: Utilise active learning to intelligently guide price changes for optimal price discovery

To overcome the lack of price dispersion in estimating price elasticity, active learning algorithms emerged as a plausible candidate.

Active Learning (AL) is a powerful machine learning technique that selects the most informative data points for training models, focusing on data that is expected to improve the model's performance the most. A typical AL framework, according to Budd et al. (2021) contains a method to evaluate the informativeness of unlabeled data points, followed by a query that selects data points to be annotated (i.e. labelled). After the allocation of new annotations, the AL framework enhances the model by either retraining using all annotated data or fine-tuning using the most recently annotated points.

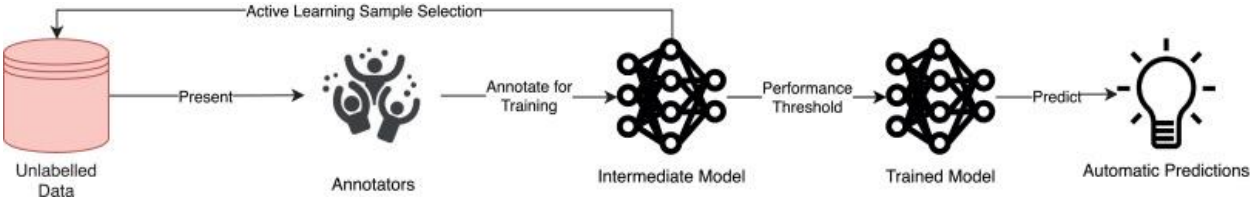

Figure 1 - AL integration steps (Budd et al., 2021)

The applications of AL techniques can be extremely versatile as they have been proven to improve the performance of various tasks. For instance, regarding automatic speech recognition, research by Riccardi and Hakkani-Tur (2005) concluded that the quantity of labeled data points required for a given word accuracy can be alleviated by over 60% compared to random sampling. Another significant use of AL is assisting the classification of multimedia such as images and videos, with multiple efforts surveyed by Wang and Hua (2011).

Likewise, AL algorithms could be leveraged to elevate the search for optimal prices at businesses, especially those reluctant to increase their price dispersion. AL techniques might learn which price points would provide the most informative demand data, then combine with a machine learning model to arrive at the desired demand curve without involving numerous costs. This

paper aims to test whether such approach is plausible and effective compared to a naïve practice of randomly choosing price points.

2. Critical literature review

2.1. Price elasticity and its estimation

It is widely recognised that the price of a certain product is one of the key drivers of the demand for that item. Thus, companies have studied this relationship and measure the responsiveness of quantity demanded to increase in price, referred to as "price elasticity" (Helmold, 2020). The formula for this metric is:

$$\varepsilon_d = \frac{Percentage\ change\ in\ quantity\ demanded}{Percentage\ change\ in\ price} = \frac{\Delta q/q}{\Delta p/p}$$

where:

ε_d = price elasticity of demand;

q = quantity demanded;

p = price.

Depending on whether the computed elasticity is higher, lower or equal to 1, the demand of a certain product can be categorised as elastic (sensitive to price changes), inelastic (non-sensitive to price changes) or unit elastic (proportionally change with price) (Prasad, 2010). This can provide initiatives for businesses to set optimal prices that balance volume and profit (Pindyck & Rubinfeld, 2013). Moreover, the relationship between price and demand is commonly assumed to be fairly linear and depicted as a flat line for elastic demand and a steep line for inelastic demand – also referred to as a "*demand curve*" (Hubbard & O'Brien, 2008; McConnell, Brue, & Flynn, 2009), based on the notion that for inelastic items, it would take a significant drop in price to improve the demanded quantities. Nevertheless, Andrews and Benzing (2010) believed such illustration could be misleading since for a steep demand curve, at a sufficiently high price, the demand might be price elastic. Henceforth, the price elasticity should be heavily dependent on the position along a linear demand curve.

The estimation of price elasticity and demand curve in real-world situations has never been a simple task. One noteworthy approach is linear regression as adopted by Bijmolt et al. (2005) where price elasticities were modelled as a linear function of the variables by applying hierarchical linear models estimated with iterative generalised least squares as inspired by Bijmolt and Pieters (2001). Overall, the study provided an estimate for the average price elasticity and extend to analyse the effect of each determinant on the elasticity's magnitude.

Many obstacles were identified by practitioners, including the large amount of aggregated data across multiple classes, resulting in abnormally high and positive elasticities (Blattberg & George, 1991), or the variability of elasticities across different price points. Hence, to address these challenges, Sui et al. (2019) introduced several econometrics models to estimate price elasticities, such as Ordinary Least Squares, Quantile Regression or Gravity Center Regression. Especially, the elimination of measurement errors (e.g. outliers, breakpoints, etc.) via simple techniques before modelling was proved to considerably boost the performance.

Attempts were also exerted to measure elasticities using data over time. For instance, Senhadji (1998) utilised several time series techniques to eliminate data non-stationarity to derive a structural demand equation and corresponding price elasticity. Furthermore, recognising that changes in the industry structure (e.g. increased competition, available discounts, etc) could render results from traditional time-series models invalid, Cracknell and Knott (1995) decided to incorporate panel cross-section data to estimate long-term elasticity, as well as regression analysis to separate the effects of promotional discounts.

2.2. Price optimisation methods

Price optimisation has been a central topic of research regarding revenue management with the goal of identifying a price strategy that maximises profit or revenue (Ito & Fujimaki, 2016). Traditionally, businesses have been determining the price of their products based on cost, competition and demand, forming three large categories of conventional pricing methods (Avlonitis & Indounas, 2005). To illustrate, cost-based pricing might involve adding elements (e.g. administrative cost, profit margin, etc.) to the estimated unit costs, referred to as the cost-plus approach (Guerreiro et al., 2012; Barusman et al., 2020), or striving to break even by setting the price so as the total revenue could recover all costs (Lovelock, 1996). Alternatively, understanding customer demand might allow companies to "*fix price according to customers' perceptions of service value to their willingness to pay*" (Calabrese & De Francesco, 2014). Nevertheless, these approaches possess distinctive limitations, such as the requirement of a profound understanding of customers' perceptions or business models for value-based methods (Töytäri et al., 2017).

Henceforth, advanced price optimisation strategies have been developed, primarily in combination with powerful machine learning algorithms. We identified that most models are associated with estimating the price-demand relationship, followed by an optimisation function. This signifies the importance of deriving an accurate demand curve, which has been proven by (Fonseca, 2017; Besbes & Zeevi, 2015; Cooper et al., 2015) to effectively produce optimal prices, although the assumed linear relationship is restrictive and simple (Constăngioară & Florian, 2018).

Examples of noteworthy price optimisation models include the basic approach proposed by Phillips (2005) which involves the formulation of linear price-response function, followed by the application of optimality conditions for unconstrained price optimisation (e.g. marginal revenue equals marginal costs). Given such basis, practitioners have incorporated sophisticated algorithms to enhance both the forecasting and optimisation performance, and overcome specific challenges. For example, to address the substitution effects between multiple products, Lee et al. (2022) identified key features of such effects, then built a demand prediction algorithm with price and critical features as inputs, utilising predictive ensemble models. The price optimisation algorithm was a constraints programming model that was subject to an increasing number of iterations and execution time. A similar problem was also studied and tackle for the e-commerce industry (Kedia et al., 2020).

For cases where demand is highly uncertain and price dispersion is limited, established research are relatively limited. Ardjmand et al. (2016) introduced a generalised price-demand structure and a modified unconscious search algorithm to obtain a set of prices for the products in each period, then determined the corresponding production and sales using a Simplex model. The approach managed to increase the average profit when implemented in a real case textile manufacturing plan. In another attempt, Ferreira et al. (2016) predicted demand for new items with no historical data using regression trees with bagging, then created an efficient optimisation model based on Linear Programming method. Nevertheless, certain limitations persist, namely high costs and random exploration for unconscious search or data intensive and slow improvement for the second model. Henceforth, this paper will contribute to the improvement of demand curve accuracy by introducing methods that can select the most informative price points, especially when obtaining data is costly.

Notably, despite the evident importance of price-demand response, price optimisation should be a multivariate analysis where various other factors must be evaluated such as competitor price, historically pricing, localised regional pricing, etc. (M.T.M Shafkhan et al., 2021), which provides a potential direction for further research.

2.3. Active learning techniques and their applications

Traditionally, machine learning research treats the learner as a passive recipient of data, thereby ignoring its powerful ability to act, gather and understand data. Thus, to address this shortcoming, AL was introduced, which studies the “closed-loop phenomenon of a learner selecting actions or making queries that influence what data are added to its training set” (Cohn et al., 1996). Critical components of an AL model include the *initial set* of labeled data, *pool of unlabeled data* to choose from, a *machine learning model* (e.g. regression, decision trees, etc.) to be trained and improved through AL, an appropriate *query strategy* to select the data instances to be labeled by and an *oracle* (e.g. human annotator) (Settles, 2011).

2.3.1. AL problem scenario

Settles (2009) provided a comprehensive explanation of how AL techniques function. First, suitable AL settings (i.e. scenarios in which the learner may query or request for labels) must be considered with the three main scenarios being *membership query synthesis*, *stream-based selective sampling*, and *pool-based active learning* (Schumann & Rehbein, 2019; Piedboeuf & Langlais, 2022; Wang et al., 2015; Kelner, 2020). For instance, under the stream-based, the learner selects and decides to query one instance at a time, whilst the pool-based approach allows the ranking of instances within a pool of unlabeled data and the most valuable points will be queried.

Given the nature of price optimisation, pool-based AL emerges as a more applicable option compared to its counterparts. The rationale includes time and cost effectiveness, and the existence of a pool of potential price points.

2.3.2. AL query strategy

After setting *when* the learner queries, it is vital to formulate a query strategy – *how* the learner selects the most informative instances. In this regard, Uncertainty sampling is considered the most popular and simplest approach (Lewis & Gale, 1994), in which an active learner queries the data points it is *least certain* to predict, thereby maximising the increase in the model's performance (Nguyen, et al., 2021; Yang et al., 2014; Raj & Bach, 2021). Other methods aiming at enhancing the performance include the Expected Model Change that chooses instances resulting in the largest change to the model (Cai, Zhang & Zhou, 2013).

Hence, within the context of this study, Uncertainty sampling could be beneficial. Another potential strategy is Greedy sampling as suggested by Wu, Lin and Huang (2019), which prioritises data points where the model's predictions are furthest from the true values. By focusing on the most problematic estimates, both methods benefit our problem as they directly concentrate on the model's accuracy improvement and avoid sophisticated calculation.

2.3.3. AL applications

Numerous studies and experiments have been published on applying AL techniques to real-world scenarios, with image/video categorisation (Joshi et al., 2009; Kapoor et al., 2007; Qi et al., 2008; Yang et al., 2017) and text/web classification (Tong & Koller, 2002; Cha Zhang & Tsuhan Chen, 2002; Miller et al., 2020) among the most common tasks. Furthermore, several papers concluded that AL could yield performance comparable to passive learning with significantly smaller training sets (Figueroa et al., 2012; Wu, 2021). Nevertheless, regarding AL algorithms' potentials to enhance pricing models and price optimisation, limited work has been performed as we failed to discover empirical research on this subject. Relevant papers include Kou et al's (2015) work on combining AL with a machine learning model to forecast electricity prices, thereby significantly reduce the computational time whilst still maintaining the desired accuracy. Although this

research did not directly address the price optimisation problem, it demonstrated AL's ability to improve the forecasting efficiency of a sophisticated machine learning algorithm, which suits our case where demand prediction is involved.

3. Methodology

The methodology section serves as the backbone of this paper where comprehensive details of our approaches, data employed, strategies selected and models developed are explained. This section starts with an overall explanation of the data being deployed for the research. Then, appropriate prediction models and AL techniques are explained to guide the optimisation of price to maximise sales or profitability. Lastly, a complete algorithm is constructed to illustrate the entire process.

3.1. Data overview

The dataset selected for this research plays a pivotal role in ensuring relevant findings and achieving the paper's targets. This section elaborates on its source, characteristics, and significance concerning the study's objectives.

3.1.1. Data source

A simulated synthetic dataset was considered appropriate for this study since our objectives focus on the effectiveness of price discovery methods in lieu of being data-centric, and it also prevents any unnecessary complexity generally observed in real-world information.

The dataset simulates weekly historical sales records of 391 products across 20 locations, spanning from 24/05/2021 to 12/06/2023. Its relevance to this research stems from its comprehensive range of variables contributing to sales prediction as well as a reasonable timeframe for analysis.

3.1.2. Data structure

Key Characteristics of the dataset can be outlined as follows:

- **Time span**: from 24/05/2021 to 12/06/2023, with weekly frequency
- **Size:** 844,560 observations on a total of 391 distinct products and 20 distinct store locations
- **Variables:** The dataset is organised in a time-series format, with each observation corresponding to sales information for a specific product at a specific location for a date. Key variables included in the dataset are:
  - **Date:** The date for each observation, with a seven-day time interval between two observations (i.e. can be considered as weekly sales).
  - **Product ID:** A unique identifier for each product.
  - **Location ID:** A unique identifier for each store location.

- **Sales Units:** The number of units sold at each observation.
- **Sales Value:** The total revenue generated from sales at each observation.
- **Return Units:** The number of products returned at each observation.
- **Selling Price:** The price at which the product was sold.
- **List Price:** The standard price of the product before any discounts.
- **Cost:** The cost incurred to produce or acquire the product.
- **Profit Margin:** The profit margin generated from selling the product.

To ensure the data quality is guaranteed, the following preliminary measures were conducted:

- **Missing values check**: There was no valid return value for the entire dataset and most entries in the "*receipted_units*" column were invalid. Thus, we decided to drop these variables in our analysis.
- **Zero values examination**: No observation was found as showing zero or negative sales, which strengthens the appropriateness of the dataset.
- **Outliers detection**: using the z-score method with a threshold of 3, we identified that the number of outliers in Price and Sales units was negligible and thus, we decided to retain such observations.

3.2. Price elasticity model

Given our objectives and the continuous nature of the dependent variable (i.e. sales), Linear Regression - a widely accepted method for estimating price elasticity, was selected to model price elasticity, which assumes a linear relationship between the input features and the target variable (Montgomery, Peck, & Vining, 2012). Since there might be multiple variables affecting the output instead of a single predictor, the relationship between variables under the Linear Regression model can be generalised as the following equation:

$\mathcal{Y} = \beta_0 + \beta_1\mathcal{X}_1 + \beta_2\mathcal{X}_2 + \cdots\cdots + \mathcal{E}$, where:

- $\mathcal{Y}$ is the dependent variable (i.e. sales units),

- $\mathcal{X}_1$, $\mathcal{X}_2$, etc. are the independent variables (e.g. prices, seasonality, etc.)

- β_0 is the intercept.

- β1, β2, etc. is the slope of each independent variable.

- ε represents the error term.

Similar to other estimating models, Linear Regression is subject to several assumptions, which were explained in detail by (Poole & O'Farrell, 1971; Casson & Farmer, 2014). Apart from the

aforementioned linear relationship between the predictors and outcome variable, critical assumptions of this model include:

- **Continuous output**: The response variable is continuous rather than categorical.
- **No measurement error**: regardless of the distribution of the covariates, there should be no random component to the predictors.
- **Conditional distribution of the error term ε has a zero mean**.
- **Homoscedasticity**: The variance of the error term ε should be constant across all levels of the independent variables, in order to ensure that the model's predictions are equally reliable across the range of data.
- **Serially independent error term**: The error terms should be independent of each other.
- **No Multicollinearity**: the predictors should be linearly independent of each other. Multicollinearity hinders the model's ability to estimate the individual effects of each independent variable.

A Linear Regression model was developed to predict sales using factors provided in the dataset. Then, AL algorithms were incorporated into this model to elevate its performance by requesting it to learn from more data. The implemented AL strategies are discussed below.

3.3. Active Learning strategy

Given the wide range of AL techniques mentioned in section 2.3, selecting and implementing a relevant strategy is pivotal in ensuring a quality outcome of the study as well as its practical applicability. In this subsection, we will explain the entire process of incorporating AL strategy to enhance a regression model's performance, including the selected AL problem scenario (i.e. when the model query new data instance) and query strategy (i.e. how to select data points to request label), followed by model retraining and evaluation.

3.3.1. AL problem scenario

As introduced in section 2.3, pool-based AL is an approach where a pool of unlabeled data is maintained, and each of the instances within the pool is evaluated by the model to determine the most impactful instance for querying (the definition of "*most impactful*" depends on specific query strategies, which will be discussed in section 3.3.2).

Under pool-based AL, as suggested by (Sugiyama & Nakajima, 2009; Liu et al., 2021), it starts with:

- A pool consisting of unlabeled samples $\mathcal{D}_u = \{x_1, x_2, \dots, x_n\}$;
- $\mathcal{Y}_i$ as the actual label for point x_i in the pool where $1 \leq i \leq n$;
- $f(x)$ as the function predicting $\mathcal{Y}$ at price x (i.e. in this case, this would be the equation of the linear regression model), and

- A set with labeled samples to develop an initial model, defined as $\mathcal{D}_l = \{(x_{m,}, \mathcal{Y}_m), \dots, (x_k, \mathcal{Y}_k)\}$, where $n \leq m \leq k$.

The first step is to build an initial regression model with function $f(x)$ by fitting the set of labeled samples $\mathcal{D}_l$. Then, the objective is to optimally select *T* number of samples from $\mathcal{D}_u$, ask the annotator to obtain their labels (i.e. $\mathcal{Y}_i$), add these samples and their outputs to $\mathcal{D}_l$, and re-train the regression model to predict future unknown data points. Ideally, AL techniques will help elevating the prediction capacity of the model using the minimum number of necessary samples (i.e. minimise *T*), thereby solving a significant obstacle in real-world situations where obtaining labels is costly. The methods to select instances from the pool $\mathcal{D}_u$ to be queried and learnt by the model are referred to as query strategies and selected approaches are explained in the next subsection.

3.3.2. AL query strategy

a. Uncertainty sampling

Under this method, price points in the pool about which the predictor is most doubtful will be queried. As such, the direct focus will be on areas (i.e. price points) where the model's confidence is lowest (most uncertain), resulting in rapid improvement in its performance and fewer experiments required.

A debatable topic of this approach is the quantification of "*uncertainty*" (Nguyen et al., 2021). In other words, it might be troublesome to use the model to mathematically evaluate each point in the pool and determine which one is most difficult to predict, particularly for regression problems as there is no prediction probabilities as in classification.

Therefore, we decided to quantify uncertainty by the variance in prediction across different model runs. To initiate multiple model runs, instead of fitting the initial linear regression model with $\mathcal{D}_l$ as mentioned in section 3.3.1, we created 10 bootstrap samples of $\mathcal{D}_l$ and fit them to create 10 separate linear regression models. Then, each model would make prediction for every point in $\mathcal{D}_u$, and for each point, calculate the prediction variance σ^2 across all 10 models (Settles, 2012). Accordingly, the data instance x^* with the highest prediction variance indicates that the models are most uncertain to estimate the output of this point, therefore, it should be selected for further training.

$$x^* = arg \max_{x_i \in \mathcal{D}_u} \sigma^2 (x_i)$$

b. Greedy sampling

Under Greedy sampling, or Greedy sampling by output, the model quantifies the impact of a data instance in the pool by measuring the distance between its predictions of the data point against the available labels (Wu et al., 2019). Alternatively expressed, the further the prediction of a data

instance is from the already available outputs, the more impactful this point will be, and therefore, the more likely it will be selected for model learning.

In a pool-based setting as defined in section 3.3.1, this strategy starts with building a regression model $f(x)$ by fitting the $(k-m)$ labeled samples in $\mathcal{D}_l$. Afterward, for each of the *n* unlabeled instances in the pool $\mathcal{D}_u$ (i.e. $\{x_i\}_{i=1}^{n}$), GS determines the distance d_{im}^{y} from the prediction of the unlabeled point (i.e. $f(x_i)$) to each of the labels in $\mathcal{D}_l$ (i.e. $\{y_m\}_m^k$) and computes the shortest gap d_i^y:

$$d_{im}^{y} = \|f(x_i) - y_m\|;\ i = 1, \dots, n;\ m = n+1, \dots, k$$
$$d_i^y = \min_m d_{im}^{y}$$

The sample with the highest distance d_i^y will be considered as most impactful and therefore, will be queried for label.

The next step involves obtaining the output labels (i.e. y_i) for the chosen data points, supplementing them to $\mathcal{D}_l$ and retrain the model to enhance its performance, which is constantly evaluated by metrics explained in section 4.3. Then, all the aforementioned steps are iterated until the desired performance or objective is reached.

c. Random selection

This is the most basic method to select instance for model learning and will be used as a benchmark to analyse any benefits of the above AL approaches.

Under random selection, data points x_i are randomly selected from the pool $\mathcal{D}_u$ without mathematical computation to evaluate their impacts. Then, the corresponding labels y_i will be obtained and other processes are similar to Uncertainty sampling and Greedy sampling.

The entire methodology described in this section can be summarised by the algorithms in ***Appendix 1*** *for ease of understanding.*

4. Research implementation

4.1. Initial data analysis

<u>Examination of data variation</u>

After understanding all key characteristics of the dataset and performing necessary checks to ensure its adequacy and reliability, we continued to analyse the variability of the data to support the research.

First, descriptive statistics of each variable such as mean, median, standard deviation and quartiles were calculated to examine the figures' range. Given that selling price and sales units

are the main variables, histograms of them across the entire dataset were plotted, which indicated substantial variability due to the wide range of products and locations.

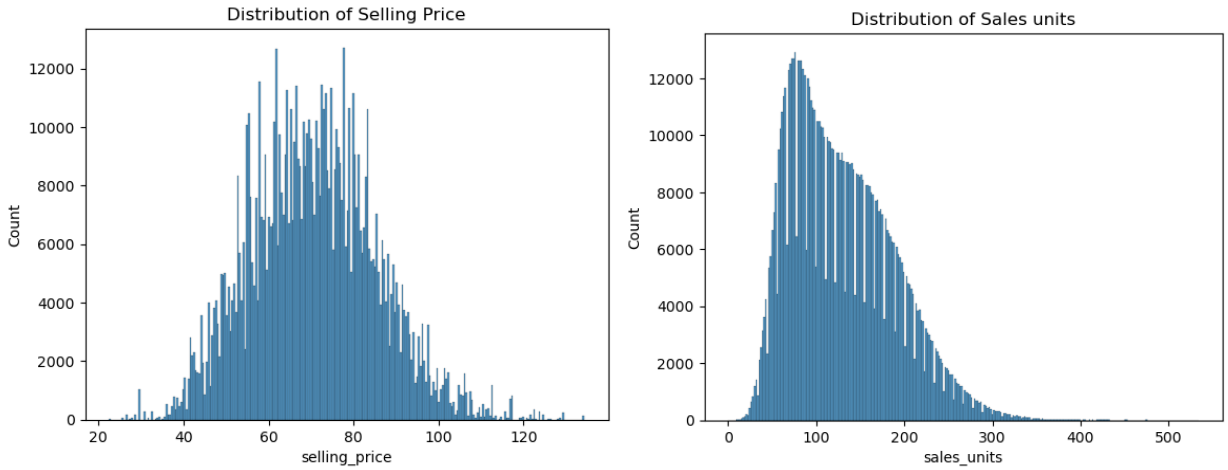

Figure 2 - Distribution of selling price and sales units across the entire dataset

Henceforth, to avoid over-complicating the study and impacts from different price elasticity of products, the price-demand relationship was modelled at a product-and-location-specific level. Accordingly, the product at the store with the most price variation and available sales data are selected for the analysis. The histogram of this subset indicated a great variability for our analysis, even for a significantly smaller dataset.

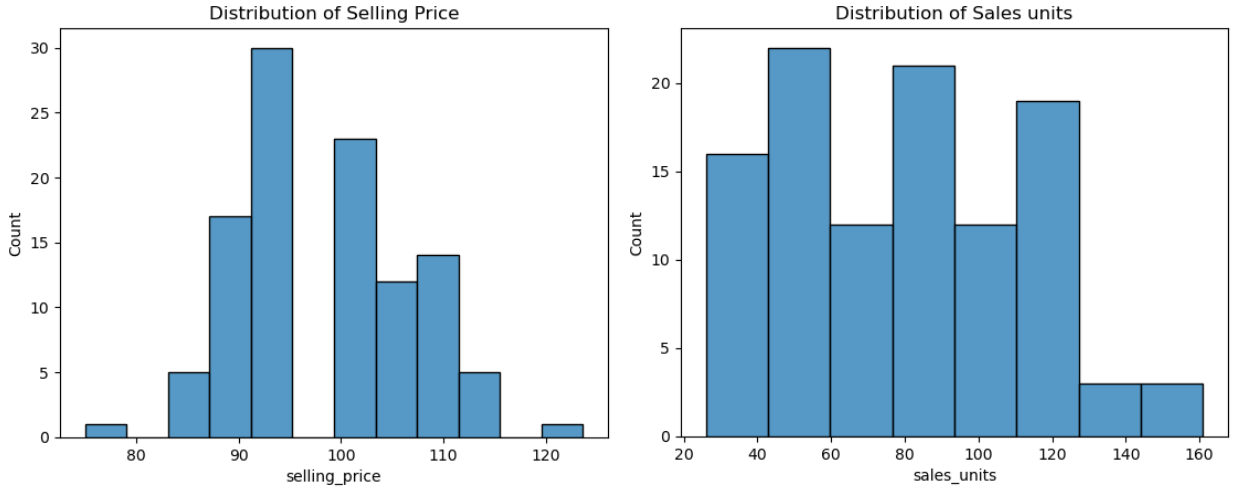

Figure 3 - Distribution of selling price and sales units for 1 product at 1 store

After extracting the appropriate sub-dataset, we plotted a time series of sales data and decomposed into trend, seasonal and residual components to identify significant patterns. Notably, both additive and multiplicative decomposition methods were implemented, and the diagrams displayed an evident monthly seasonal component. The same procedures were implemented for other products and locations, which suggested such seasonal component was

consistent and should be considered as a factor in predicting sales. Conversely, an explicit trend was not visible from any decomposition plot for most products. The diagrams for our chosen product and store can be referred to as follows.

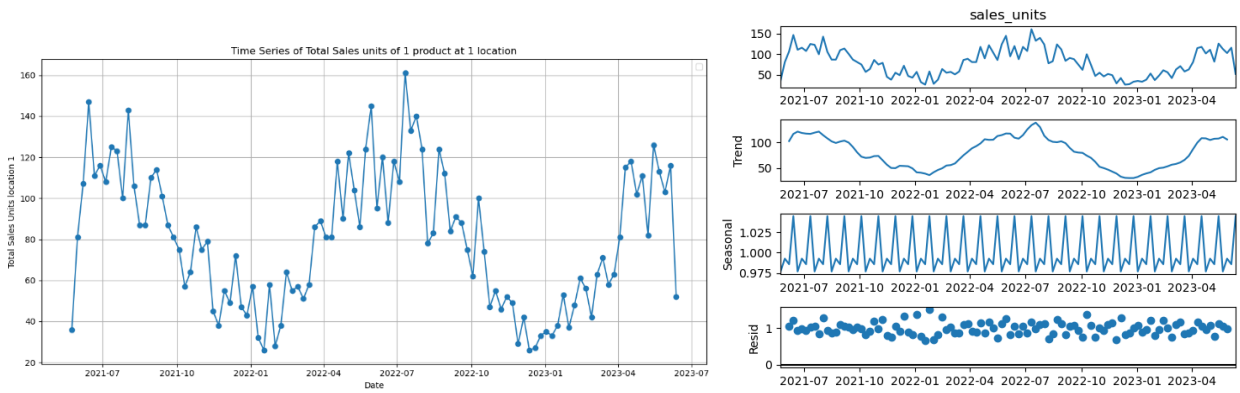

Figure 4 - Time series decomposition

<u>Baseline model and product elasticity</u>

The baseline model was developed as a linear regression model based on the selected sub-dataset. To facilitate the modeling process, the data were first preprocessed as follows:

- **Log-Transformation**: Both the selling price and sales quantities were log-transformed to stabilise variance and normalise the distribution.
- **Seasonality factor**: The date variable was decomposed into a seasonality index using multiple month dummy variables to capture periodic fluctuations in demand as identified above.
- **Feature Selection**: The final feature set included the log-transformed selling price and seasonality dummy variables (months) to estimate the dependent variable being the log-transformed sales units.

The subset was then split into train, pool and test portions in chronological order (i.e. train set was the first 20% of the time series, pool set was the next 60% and test set accounted for the remaining). The rationale is that as we were unable to obtain real-time sales for different price points, a pool with actual (historical) output could be a substitute to build the model under analysis.

To develop the baseline model, the train set was fitted using the linear regression method with the aforementioned features and target variable. Accordingly, the coefficient associated with the price variable in this model represents the price elasticity of demand for each product since it indicates the expected percentage change in sales resulting from one percent change in price.

Finally, the baseline model was evaluated using statistical metrics, which displayed a relatively modest R-squared (i.e. 0.2487) and moderate levels of RMSE and MAPE. Hence, there existed potentials for improvement by implementing AL techniques.

4.2. Active Learning cycle

4.2.1. Initial Price Setting

The AL process began with the preliminary model fitted using the train subset described in Section 4.1. This step is crucial as it sets the baseline for further iterations, allowing the model to establish an initial understanding of price elasticity for this particular product and store location. Based on such understanding, it then selected the price point within the test set that maximised the expected profitability, which is the ultimate objective of our modelling.

At this stage, the model's ability to accurately predict the impact of price changes was relatively limited due to the modest training data. As such, the determined optimal price and other metrics might not be positive and would serve as a benchmark for the results from refined models.

4.2.2. Selecting informative data points

The core of the AL Cycle involves systematically improving the model through the strategic selection of new data points within the separate pool. At the beginning of each iteration, the most updated model was used to predict the target variables (i.e. sales quantity) for the remaining unlabeled data in the pool. Then, depending on the query strategy, the model would select instances to learn from the pool based on:

(i) Uncertainty sampling: how uncertain the model is when estimating. The degree of uncertainty was quantified using prediction variance as explained in Section 3.3.2. The instance with the highest variance in predictions is considered the most uncertain and, therefore, the most effective for further model refining.

(ii) Greedy sampling: distance from the predicted figures to labelled data. The estimated sales of each instance in the pool were compared to every output in the labelled set, and the shortest gap was selected as the final distance. Then, the data point with the maximum final distance were considered as most informative.

Random selection of data points in the pool, regardless of their informativeness, to update the model in each iteration was also implemented for comparison purposes.

4.2.3. Updating the model with new data points

After identifying the data point to learn from, the corresponding true label (i.e. actual sales data) was added to the existing labelled dataset, expanding the training set. This enabled the model to continuously improve as it underwent retraining using more data after each iteration (including original training data and newly incorporated points).

The purpose of retraining is to refine the model's understanding of the relationship between sales and input features. Each new data point contributed to a more accurate estimation of price elasticity, as the model was exposed to a broader range of scenarios and price points, ultimately leading to more reliable estimates and optimisation capabilities. To ensure this objective is achieved, we evaluated the model after each iteration and utilised visualisation tools to analyse the impacts using certain metrics, which will be discussed in later sections.

4.2.4. Repeating the cycle for enhanced model accuracy

The last step is to repeat the aforementioned processes to create an AL loop for the model. The number of iterations could be predetermined based on the desired level of key metrics, or the size of the data pool. Alternatively, the cycle might continue until the level of performance reaches a plateau, or the addition of data points provides no benefits to the model.

4.3. Evaluation metrics

Since different algorithms selected different data points to learn, the remaining unlabeled samples in the pool after each iteration were different for each method. Hence, to establish consistency in the evaluation process, a test subset was separated from the original dataset and remained intact throughout the cycle.

4.3.1. Model performance

Model performance was assessed by computing the accuracy of demand predictions with the following metrics, assuming *n* is the number of values in the test dataset:

- **Mean Absolute Percentage Error (MAPE):** measures the average absolute percentage difference between the predicted and actual sales.

$$MAPE = \frac{100}{n} \times \sum_{i=1}^{n} \left| \frac{Actual\ sales - \ Predicted\ sales}{Actual\ sales} \right|$$

  Thus, a lower MAPE indicates higher accuracy, which can be interpreted that, on average, the model's predictions deviate less from the actual figures. MAPE is a common method to determine forecasting precision due to its "reliability, unit-free measure, ease of interpretation, clarity of presentation" (Montaño Moreno et al, 2013).

- **Root Mean Squared Error (RMSE):** calculates the square root of the averaged squared error, with equal weights to all the errors:

$$RMSE = \sqrt{\frac{1}{n} \times \sum_{i=1}^{n} (Actual\ sales - Predicted\ sales)^2}$$

RMSE penalises large errors in sales predictions compared to MAPE, although there might be a potential difficulty in interpretation (Goodwin, 2018, p.42).

- **Coefficient of determination (R-squared):** represents the proportion of the variance in the dependent variable (sales) that is explained by the independent variables (e.g., price, seasonality). An R-squared value closer to 1 indicates that the model explains a high proportion of the variance in sales, suggesting a strong price elasticity.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(Actual\ sales - Predicted\ sales)^2}{\sum_{i=1}^{n}(Actual\ sales - Average\ sales)^2}$$

4.3.2. Efficiency

In the context of this study, efficiency refers to the time needed to achieve satisfactory model performance. In other words, the AL approaches are considered as more efficient compared to random data selection if they require fewer iterations or unlabeled data points to reach the same accuracy or maximum profitability, hence saving valuable resources for companies in real-world scenarios.

Accordingly, as we added one data point for every instance, the proxy for efficiency would be the number of iterations until statistical metrics e.g. R-squared, RMSE and MAPE) displays a plateau or there is no significant rise in the optimal profits.

4.3.3. Optimisation capability

Optimisation refers to the model's ability to select the optimal price point that maximises either predicted sales or profitability whilst minimising the gap between estimated and actual figures. After each iteration of the AL loop, the updated model was used to project sales for every price point in the test set, and then select the optimal price point based on the specified objective:

(i) **Maximise demand:** choose prices that would generate the most predicted sales. Henceforth, AL approaches were evaluated at each iteration based on the level of demand cultivated by setting the optimal price.

(ii) **Maximise profitability:** businesses might place greater focus on profits generated compared to sales as they reflect the financial impacts of the model and AL strategies. Thus, in each iteration, the model also calculated the expected profits for every potential price within the test set, using the following formula:

$$Expected\ profit = Expected\ sales\ \times (Selling\ price - Unit\ cost)$$

The price point at which the business can capture the highest projected profits was selected.

Once the optimal price is selected, predicted sales and actual output (i.e. from the test set) are plotted to visualise the estimate gap. A smaller gap indicates that the model's predictions are closely aligned with reality, which is crucial for making reliable pricing decisions, thereby enhancing the model's utility in real-world scenarios.

5. Results and Discussion

5.1. Comparative analysis of performance

Having implemented the AL-guided and random price discovery approaches described in Section 4, we obtained and compared the results among the models to analyse whether AL could facilitate the price optimisation process at businesses. The results were separated into three categories depending on the metrics used as mentioned in Section 4.3.

5.1.1. Prediction performance

We measured the MAPE, RMSE and R-squared of the updated model after each iteration on the test dataset to evaluate any benefits or improvements on the model derived from adding one data point to the training set. To enhance the reliability of our results, we also implemented the entire procedures for the dataset of another store. The performance metrics across the three implemented approaches were plotted as follows for comparison purposes.

- **MAPE:** The percentage error of Uncertainty sampling was significantly lower than the other two methods from iteration 1 to 25. From iteration 30 onward, Greedy sampling outperformed the rest at Store 1, however this result was not consistent as this method performed poorly at Store 2.

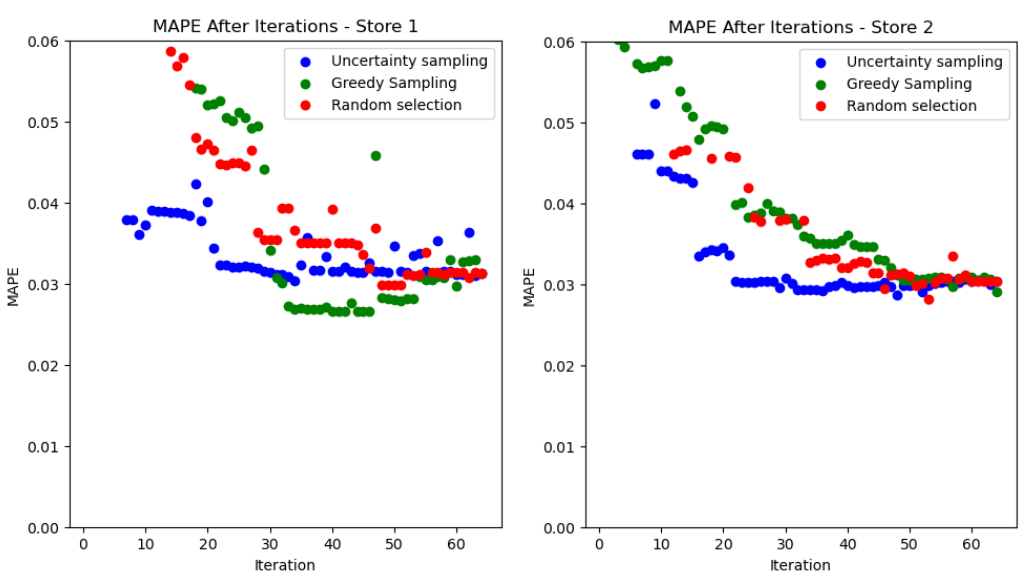

Figure 5 - MAPE result plots

- **RMSE:** The results were relatively similar to MAPE, with Uncertainty sampling generating the best metrics in early iterations. The performance of other methods reached Uncertainty sampling's levels after nearly 30 iterations and remained close to each other until convergence.

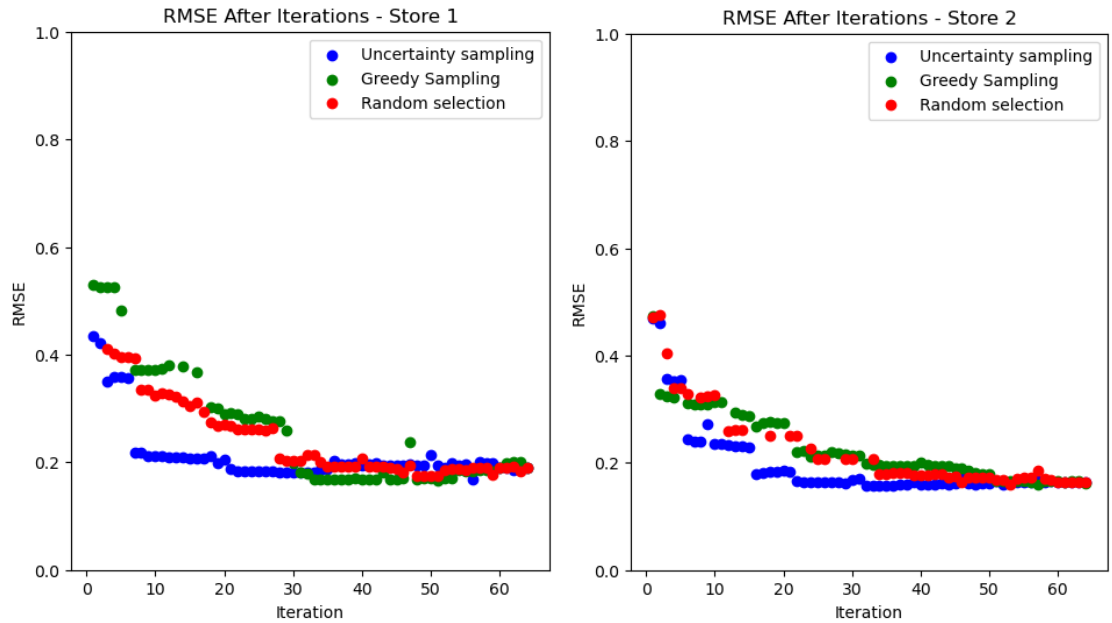

Figure 6 - RMSE result plots

- **R-squared:** The results were consistent with MAPE and RMSE, suggesting the dominance of Uncertainty sampling over the others.

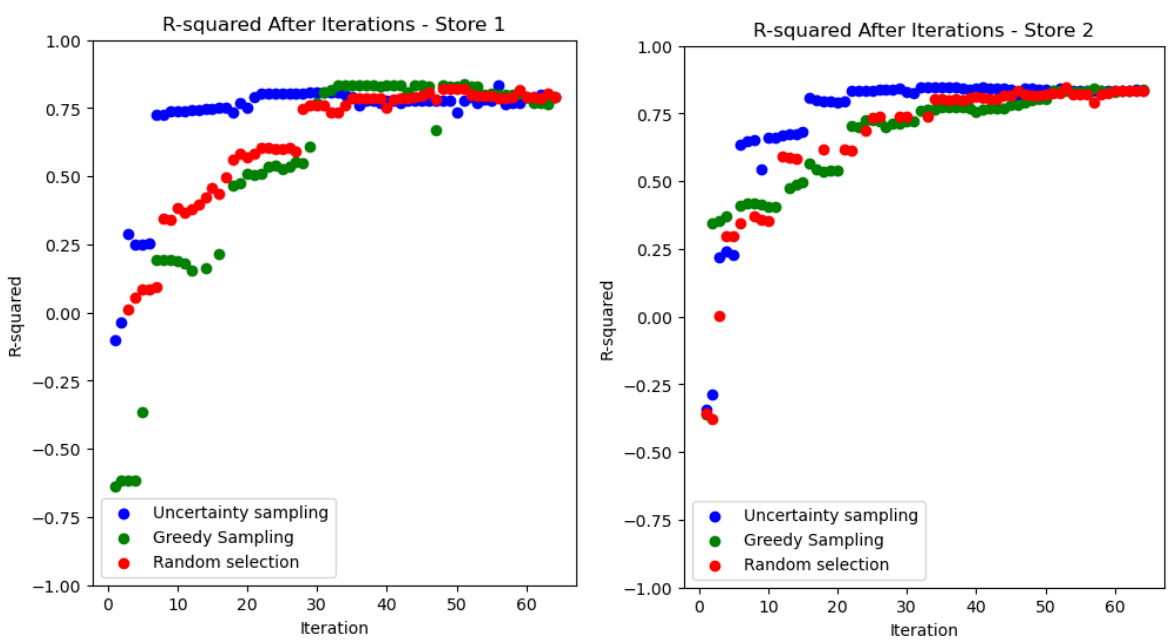

Figure 7 - R-squared result plots

Overall, Uncertainty sampling's performance was superior to Random selection as well as another AL-guided method (i.e. Greedy sampling) at all three metrics, until all methods converged (i.e. after all iterations, which is 64 in this case – the number of samples in the pool), which was consistent for both stores. Henceforth, we can expect this AL technique to enhance a model's predicting capabilities significantly faster than randomly selecting data points for model learning. Nevertheless, Greedy sampling did not prove to be more effective than Random selection since their performances vary after each iteration. Thus, selecting the appropriate AL method for each dataset, problem, situation, etc. is necessary since Random selection can still outperform certain AL techniques in our case.

5.1.2. Efficiency

As observed from the plots of calculated statistics (i.e. Figure 6, 7 and 8), Uncertainty sampling guided the model to reach a significant level of prediction performance the quickest among the three models. For instance, the MAPE of the model with Uncertainty sampling reached a plateau after roughly 22 iterations for Store 1, whilst the figures for Greedy sampling and Random selection are 31 and 28. A similar pattern could be witnessed for both RMSE and R-squared as the metrics obtained from Uncertainty sampling accelerated from the beginning and reached a constantly high level earlier than its two counterparts. This signaled that data points selected by Uncertainty sampling enhanced our model significantly considerably more rapidly compared to other approaches, and the necessary number of iterations could be reduced by over 30% compared to other approaches.

Comparing Greedy sampling and Random selection in terms of efficiency, we identified a lack of supremacy of the AL-guided algorithm since the model using Greedy sampling failed to achieve a great level of accuracy with lower number of price experiments.

5.1.3. Price optimisation capabilities

As described in Section 4.3.3, we aimed to measure the price optimisation ability of the model by requesting it to select a price in the test set that maximises either predicted sales or profitability after each model update. Then, the model would be considered as sufficient when it selects the price that generates considerable sales or profits, and simultaneously limiting the discrepancy between predicted and actual figures (in this case, actual sales and profits are available in the test set for comparison purposes). Hence, the results can be plotted as a diagram of maximum sales and profits at each iteration, across all three models as follows:

- **Sales maximisation:** the model selected price points in the pool that generated the maximum predicted sales quantity at each iteration. The highest estimated demand after each update for each model is graphed below:

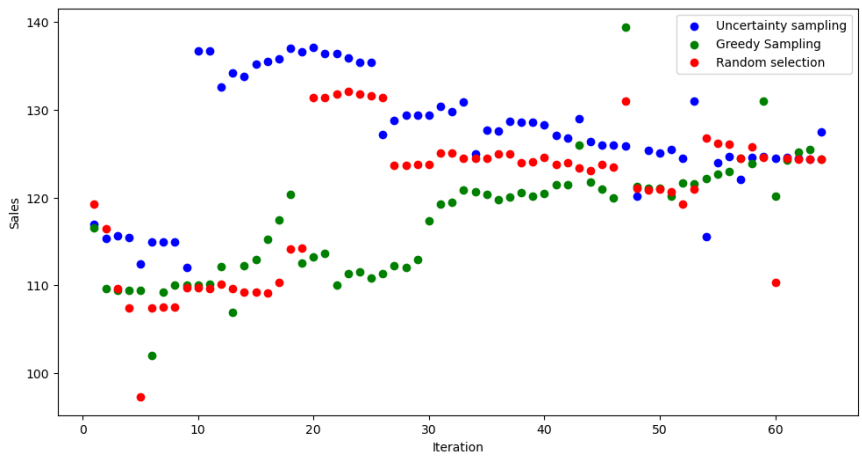

Figure 8 - Maximum expected sales quantity at each iteration

- **Profit maximisation:** the model prioritised profitability and chose prices that maximised expected profit instead of merely sales quantity. The corresponding plot is as follows:

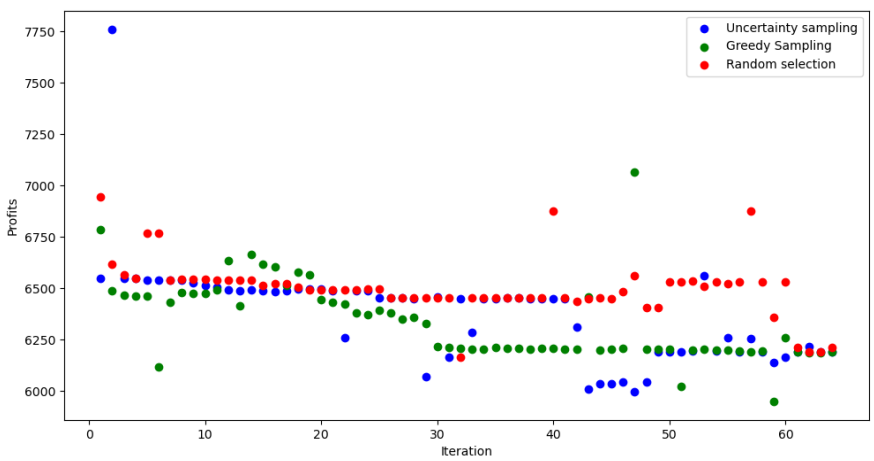

Figure 9 - Maximum expected profit at each iteration

It can be observed that Uncertainty sampling predicted the highest maximum demand amongst all the models in most iterations, suggesting that it might be an effective tool if businesses prioritise demand. In other words, Uncertainty sampling can help companies select prices that would generate significant product demand with a small number of price elasticity experiments (i.e. iterations). This is supported by its strong performance in prediction as analysed in Section 5.1.1, which indicates the estimated maximum sales is reasonable and achievable. We also visualised the predicted sales by the model with Uncertainty sampling on the test set against actual sales to verify the accuracy. Please note that the graph below depicts only 15 iterations to avoid over-complication.

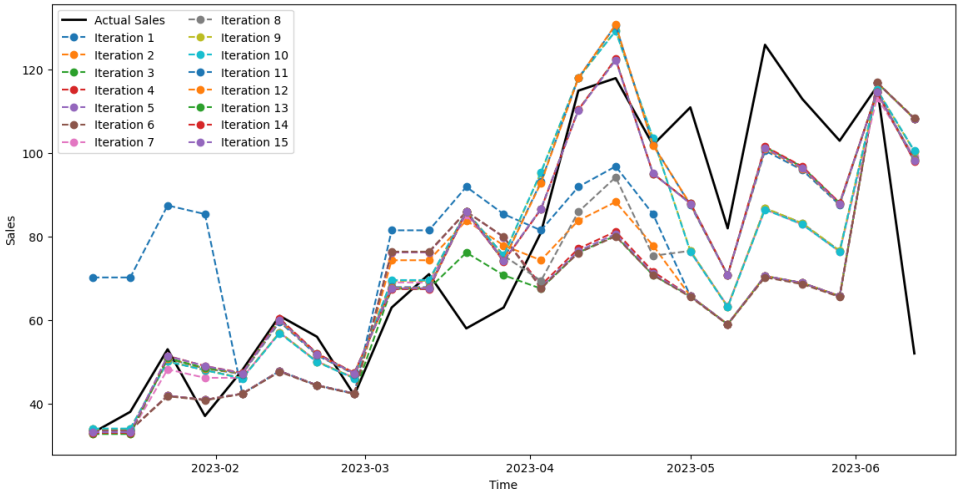

Figure 10 - Predicted vs Actual sales - Uncertainty sampling

However, Uncertainty sampling's optimisation performance did not outshine Random selection based on maximum expected profitability, indicating the lack of effectiveness of AL techniques in this case. Additionally, Greedy sampling failed to produce better outcomes compared to the non-AL approach for both sales and profit maximisation, similar to the situation with prediction metrics. The comparison plots between actual and predicted sales for Greedy sampling and Random selection can be referred to below.

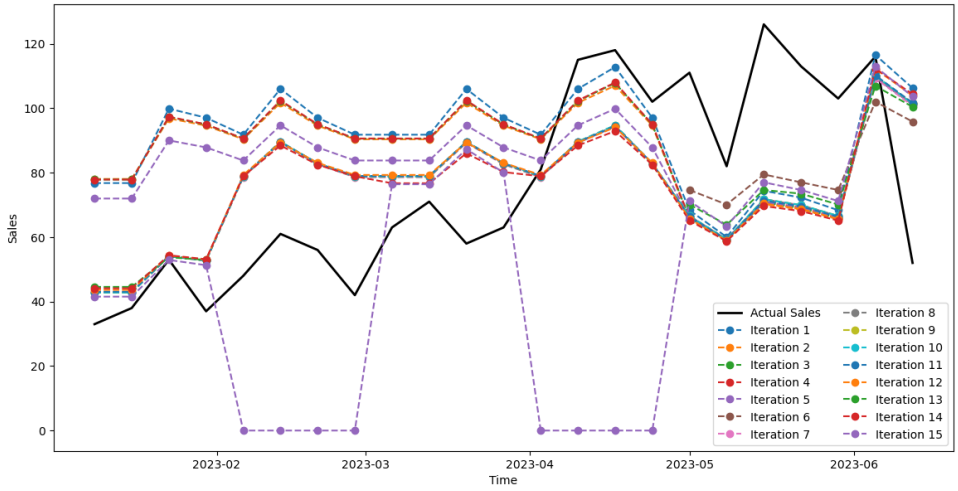

Figure 11 - Predicted vs Actual sales - Greedy sampling

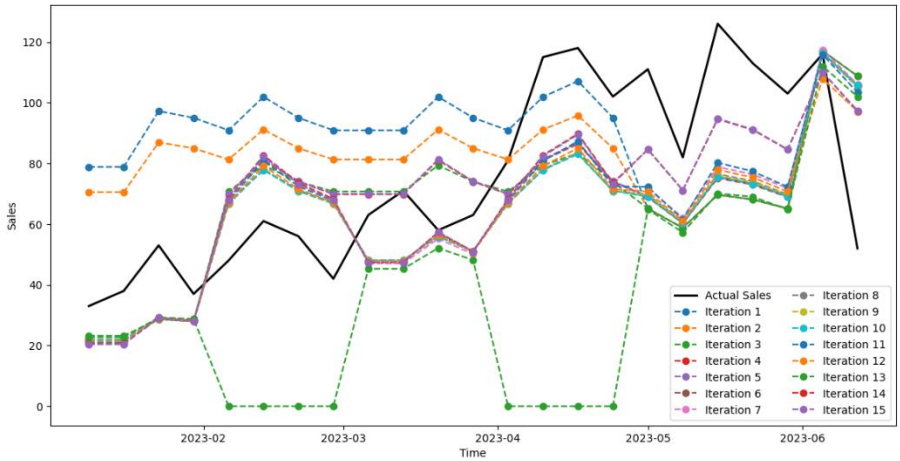

Figure 12 - Predicted vs Actual sales - Random selection

Hence, the results further strengthen the importance of choosing suitable AL techniques based on different scenarios since the mere implementation of AL algorithms does not guarantee a greater performance for businesses.

5.2. Findings and discussion

5.2.1. Insights into the effectiveness of AL in reducing data collection efforts

The above study's results cement the practicability of AL techniques in effectively reducing the needs of data collection for developing an accurate regression model. This outcome is particularly important in scenarios where obtaining labeled data is costly or time-consuming, which is, in fact, common in the business world.

For our case, traditionally, building a robust machine learning model capable of predicting sales based on price and other factors (i.e. price elasticity modelling) involves setting numerous different prices and gathering their respective sales for the model to learn until it is sufficiently strong. This approach represents the aforementioned Random selection method employed where price points are chosen randomly. However, with the integration of AL techniques (i.e. Uncertainty sampling and Greedy sampling), the model could be selective about which data point it should learn from, meaning that only a fraction of the total available data needed to be labeled and used in training. As a result, given that the performance metrics – such as RMSE, MAPE and R-squared – of an AL-guided model can rival those of models trained on larger datasets (e.g. R-squared of a model with Uncertainty sampling after only 7 iterations was approximately equal to that of Random selection after 25 iterations!), it was visibly demonstrated that it is possible to achieve a high-performing model with significantly fewer data points (i.e. the reduction in data points required could amount to over 30% as observed from the prediction metric plots).

Hence, the overall cost, time, and effort associated with data collection would be substantially reduced. The reduced need for comprehensive data collection allows for improved resources

allocation and continuous refinement, which is particularly advantageous in fast-moving markets where conditions can change rapidly. AL-associated approach also mitigated the diminishing returns commonly identified when supplementing data to an already well-trained model, as it only focused on data points that directly contributed to model refinement.

Nevertheless, focus should be placed on selecting the suitable AL techniques depending on the available dataset, business scenarios, etc. since our results also suggested that unlike Uncertainty sampling, another AL technique – Greedy sampling did not succeed in outshining its simpler counterpart, both in terms of predicting accuracy and efficiency. Hence, this might result in overcomplicating the model or waste of resources without gaining any benefits.

5.2.2. Implications for business decision-making and pricing strategies

The findings from this study carry crucial implications for business planning, particularly in the context of optimising demand and profit through data-driven approaches. One of the most compelling insights is the potential for companies to develop their demand-price relationship models using fewer data points without compromising accuracy through the integration of AL techniques.

As they were proved to be superior to the traditional approach (i.e. Random selection), such state-of-the-art methods will be increasingly valuable in scenarios where demand data is expensive, historical price variation is limited and pricing decisions must be flexible and responsive. For instance, companies wanting to understand consumer behaviour related to their existing products can rely on the AL-guided models implemented in this study to considerably reduce the costs and time of obtaining demand levels, compared to simply selecting price points to test in an aimless manner. As such, by arriving at an accurate price elasticity curve with fewer efforts, companies can set price that maximise their sales or profitability earlier, thereby avoid sacrificing any profit due to mispricing before the optimal price is identified.

Furthermore, the study underscores the value of continuous AL integration into the pricing strategy development process. By continuously suggesting informative price points to test and updating the model, AL can help businesses maintain an up-to-date understanding of price elasticity and consumer preferences. This ongoing refinement process ensures that pricing decisions are based on the latest market conditions, reducing the risk of mispricing.

For businesses, this enables pricing strategies to be more finely tuned to achieve various objectives in addition to demand and profit maximisation. The iterative nature of AL allows for experimentation with different pricing scenarios, providing a framework for testing and validating strategies before full-scale implementation. This alleviates the probability of adverse outcomes from price adjustments and enhances the ability to predict the financial implications of pricing decisions with greater confidence.

6. Conclusion and Recommendations

6.1. Summary of findings

This study explored the application of AL techniques in the context of price elasticity modeling and price optimisation, offering key findings that can significantly contribute to both academic research and practical business strategies. The primary objective was to investigate the effectiveness of AL techniques in alleviating the amount of data required to build robust price elasticity models, thereby minimising demand data collection efforts while maintaining prediction accuracy. A simpler non-AL approach was also implemented for a more detailed assessment of the impacts generated by AL methods through comparative analysis.

We utilised a synthesized dataset of a retailer with price-demand data on multiple products at different locations. For simplicity, a demand prediction model was developed to analyse the behaviour of a single product at a single store, and AL algorithms were implemented to help boost the model's performance whilst minimising the amount of additional data needed. Then, the model was assessed on its ability to forecast demand, reduce data collection and select prices to maximise sales or profitability.

The findings from the study confirmed that AL techniques, particularly Uncertainty sampling, could be highly effective in reducing data requirements for training accurate models. By selectively choosing data points that contribute most to improving model predictions, AL enabled the creation of models that required significantly fewer data points than those developed through traditional methods, such as Random selection of price points. For instance, the study demonstrated that the model using Uncertainty sampling requires less than 30% data points to reach a performance that was comparable to its counterparts.

This reduction in data collection translates directly to cost and time savings, which are critical considerations in the business world, where resources are generally limited. Hence, AL could be regarded as a powerful tool for businesses to help understand customer behaviour and price elasticity of products, allowing for quicker and more efficient responses to changing market conditions. Furthermore, with effective price-demand predictive models, companies will be able to set optimal prices to achieve certain outcomes (i.e. sales and profit optimisation in this case), thus cementing the potential benefits of AL in improving the overall competitiveness and profitability of a business.

The study also provided valuable insights into the limitations of different AL strategies. While Uncertainty sampling proved highly effective, Greedy sampling did not outperform the simpler Random selection approach. This suggests that while AL can bring substantial benefits, the selection of the AL strategy must be tailored to the specific characteristics of the data and the

business objectives. This nuance highlights the importance of understanding the underlying mechanisms of AL techniques and their suitability for different scenarios.

6.2. Recommendations

First, companies should consider integrating AL into their pricing discovery processes to alleviate the costs and time associated with increasing price variation. By focusing on the most informative prices, corporates, especially those struggling with obtaining demand data, can develop accurate price elasticity curves with fewer efforts (i.e. over 30% reduction in this case), allowing for more efficient allocation of resources across other strategic initiatives. For instance, with an accurate elasticity model, retailers can better adjust their inventory or promotion campaigns to customer demand, resulting in reduced excess inventory and stockouts. Ultimately, the overall operational efficiency will be raised without sacrificing substantial profitability due to price experimentation.

Another notable suggestion is AL integration should be a continuous process instead of a one-off project. As market conditions, consumer preferences, and competitive landscapes evolve, the models need to be regularly updated with new data. Hence, businesses should establish a robust system for continuously monitoring model performance and implementing necessary adjustments. This guarantees that pricing strategies remain relevant and effective over time, adapting to shifts in the market and maintaining a competitive edge.

For future research, one limitation of our work is the lack of changes in cost structure. This study' calculated expected profitability using the unit cost of the product, which is relatively constant based on the given dataset. Henceforth, to better replicate real-world business problems, further studies might add other factors into the cost of the products, including fixed cost or varying unit costs to test the effectiveness of AL strategies in profit maximisation.

Another important direction for future studies is the inclusion of other factors impacting sales in practice, such as competitor's price or distance to the closet competitors, resulting in improved prediction accuracy and realistic impacts of AL. Nevertheless, as more features are added, the integration of AL techniques will require more attention since the pool of data points must include all these features.

Lastly, further research could inspect the impact of AL on optimising price for newly introduced offerings, which is excluded in this paper. Since there is no historical price-demand data, a potential approach could be forming groups of products with identical characteristics or demand. Then, we can use past data of the group that most resembles the new item to develop the initial algorithm, followed by the application of AL strategies. This will add another dimension to the practicability of AL approaches and potentially discover a powerful solution to this common problem in the business world.

Appendix – Algorithms of methodology

Table 1 - Algorithm for Uncertainty sampling

Algorithm 1: The Uncertainty sampling approach

Input:

- A pool of n unlabeled samples $\mathcal{D}_u = \{x_i\}_{i=1}^{n}$, where $1 \leq i \leq n$;
- A set of labeled samples $\mathcal{D}_l = \{x_m, y_m\}_m^k$, where n $\leq m \leq k$.
- N: number of iterations, where $1 \leq N \leq n$

Output:

- Linear regression model $f(x)$ predicting y at price x;
- Updated set of labeled data $\mathcal{D}_l$

#Develop initial model

1. Train the model $f(x)$ by fitting the samples in $\mathcal{D}_l$

#Create multiple models to measure uncertainty

2. Generate B bootstrap samples of $\mathcal{D}_l$
3. Train B different models $\{f(x_1), f(x_2), \dots, f(x_B)\}$ on each of the bootstrap sample.

#Active learning iteration

for $i = 1, \dots, N$ **do**

#Measure uncertainty

for $b = 1, \dots, B$ **do**

4. For each point x_i in $\mathcal{D}_u$, obtain the prediction $\widehat{y_b}(x_i)$ from each model created from bootstrap samples of $\mathcal{D}_l$

5. Compute the mean prediction across all bootstrap models:

$$\overline{y}(x_i) = \frac{1}{B} \sum_{b=1}^{B} \widehat{y_b}(x_i)$$

6. Calculate the prediction variance:

$$\sigma^2(x_i) = \frac{1}{B} \sum_{b=1}^{B} (\widehat{y_b}(x_i) - \overline{y}(x_i))^2$$

#Select instance to query

7. Select the data point x^* from $\mathcal{D}_u$ with the highest variance $\sigma^2(x_i)$

#Query annotator

8. Obtain the true label y^* for the selected data point (e.g. actual sales).

#Update the training set

9. Add (x^*, y^*) to the labeled set $\mathcal{D}_l$
10. Remove x^* from $\mathcal{D}_u$

#Re-train the model

11. Repeat from step 1 with the updated $\mathcal{D}_l$ until N number of iterations is reached.

end

Table 2 - Algorithm for Greedy sampling

Algorithm 2: The Greedy sampling approach

Input:
- A pool of n unlabeled samples $\mathcal{D}_u = \{x_i\}_{i=1}^{n}$, where $1 \leq i \leq n$;
- A set of labeled samples $\mathcal{D}_l = \{x_m, y_m\}_m^k$, where n $\leq m \leq k$.
- N: number of iterations, where $1 \leq N \leq n$

Output:
- Linear regression model $f(x)$ predicting y at price x;
- Updated set of labeled data $\mathcal{D}_l$

#Develop initial model

1. Train the model $f(x)$ by fitting the samples in $\mathcal{D}_l$

#Active learning iteration

for $i = 1, \dots, N$ **do**

#Measure distance

2. Obtain the prediction $f(x_i)$ using the model from Step 1.
3. Compute the distance d_{im}^y from the prediction to each of the labels in $\mathcal{D}_l$ (i.e. $\{y_m\}_m^k$):

$$d_{im}^y = \|f(x_i) - y_m\|;\ m = n+1, \dots, k$$

4. Determine the shortest gap d_i^y:

$$d_i^y = \min_m d_{im}^y$$

#Select instance to query

5. Select the data point x^* from $\mathcal{D}_u$ with the highest distance d_i^y

#Query annotator

6. Obtain the true label y^* for the selected data point (e.g. actual sales)

#Update the training set

7. Add (x^*, y^*) to the labeled set $\mathcal{D}_l$
8. Remove x^* from $\mathcal{D}_u$

#Re-train the model

9. Repeat from step 1 with the updated $\mathcal{D}_l$ until N number of iterations is reached.

end

Table 3 - Algorithm for Random Selection

Algorithm 3: The Random Selection approach

Input:
- A pool of n unlabeled samples $\mathcal{D}_u = \{x_i\}_{i=1}^{n}$, where $1 \leq i \leq n$;
- A set of labeled samples $\mathcal{D}_l = \{x_m, y_m\}_m^k$, where n $\leq m \leq k$.
- N: number of iterations, where $1 \leq N \leq n$

Output:
- Linear regression model $f(x)$ predicting y at price x;
- Updated set of labeled data $\mathcal{D}_l$

#Develop initial model

1. Train the model $f(x)$ by fitting the samples in $\mathcal{D}_l$
#Select instance to query
2. Randomly select a data point x^* from $\mathcal{D}_u$
#Query annotator
3. Obtain the true label y^* for the selected data point (e.g. actual sales)
#Update the training set
4. Add (x^*, y^*) to the labeled set $\mathcal{D}_l$
5. Remove x^* from $\mathcal{D}_u$
#Re-train the model
6. Repeat from step 1 with the updated $\mathcal{D}_l$ until N number of iterations is reached.

end

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